

note

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1 Problem

$$\mathrm{Tr} A(\log A + B - \log B) \leq \frac{1}{s} \mathrm{Tr} (B^{\frac{-s}{s+1}} A B^{\frac{-s}{s+1}})^{s+1}, \forall s \in [0, 1]$$

Restate the inequality by $\alpha = s + 1$, we can get

$$\mathrm{Tr} A(\log A + B - \log B) \leq \frac{1}{\alpha - 1} \mathrm{Tr} (B^{\frac{1-\alpha}{2\alpha}} A B^{\frac{1-\alpha}{2\alpha}})^\alpha, \forall \alpha \in [1, 2]$$

2 Properties

Definition 2.1. *The von Neumann relative entropy $D(A|B)$ is defined by*

$$D(A|B) = \mathrm{Tr} A(\log A - \log B)$$

Definition 2.2. *The sandwiched quasi-relative entropy $\tilde{Q}_\alpha(A|B)$ is defined by*

$$\tilde{Q}_\alpha(A|B) = \mathrm{Tr} (B^{\frac{1-\alpha}{2\alpha}} A B^{\frac{1-\alpha}{2\alpha}})^\alpha, \forall \alpha \in (1, \infty)$$

Definition 2.3. *The α -sandwiched Renyi relative entropy $\tilde{D}_\alpha(A|B)$ is defined by*

$$\tilde{D}_\alpha(A|B) = \frac{1}{\alpha - 1} \log \tilde{Q}_\alpha(A|B)$$

Some properties of sandwiched quasi-relative entropy $\tilde{Q}_\alpha(A|B)$

1. $\lim_{\alpha \rightarrow 1} \frac{1}{\alpha - 1} \log \tilde{Q}_\alpha(A|B) = D(A|B)$ ([2] Proposition 11)
2. If $\alpha < \beta$, then $\tilde{D}_\alpha(A|B) < \tilde{D}_\beta(A|B)$ ([3] Proposition 3.1.2)
3. $\tilde{Q}_\alpha(A|B)$ is differentiable in α ([1] Proposition 15)
4. $\tilde{D}_\alpha(A|B) \geq 0$ and equality holds when $A = B$ ([1] Theorem 3)

Corollary 2.0.1. *The monotonicity of $\tilde{D}_\alpha$ in α implies monotonicity of $\tilde{Q}_\alpha$ for $\alpha > 1$*

Proof. Let $1 < \alpha < \beta$.

$$\begin{aligned}\frac{1}{\alpha-1} \log \tilde{Q}_\alpha(A|B) &< \frac{1}{\beta-1} \log \tilde{Q}_\beta(A|B) \\ \log \tilde{Q}_\alpha(A|B) &< \frac{\alpha-1}{\beta-1} \log \tilde{Q}_\beta(A|B) < \log \tilde{Q}_\beta(A|B) \\ \tilde{Q}_\alpha(A|B) &< \tilde{Q}_\beta(A|B)\end{aligned}$$

□

Corollary 2.0.2. $\lim_{\alpha \rightarrow 1} \tilde{Q}_\alpha(A|B) = \tilde{Q}_1(A|B) = \text{Tr } A = 1$

3 Main Proof

We will first check the claim that

Claim 1.

$$\text{Tr } A \left(\log \frac{A+B}{2} - \log B \right) \leq \frac{1}{\alpha-1} \tilde{Q}_\alpha(A|B), \forall \alpha \in [1, 2]$$

Proposition 1. *Given density operators A, B , we have*

$$\text{Tr } A \left(\log \frac{A+B}{2} - \log B \right) \leq D(A|B)$$

and equality holds if and only if $A = B$

Proof. Observe $D(A|B) - \text{Tr } A \left(\log \frac{A+B}{2} - \log B \right) = D(A|\frac{A+B}{2}) \geq 0$ □

Using the proposition 1 and property 1, we can reduced our problem to

$$\lim_{\alpha \rightarrow 1} \frac{1}{\alpha-1} \log \tilde{Q}_\alpha(A|B) \leq \frac{1}{\beta-1} \tilde{Q}_\beta(A|B), \forall \beta \in [1, 2]$$

Proposition 2. *Given density operators A, B ,*

$$\frac{1}{\alpha-1} \tilde{Q}_\alpha(A|B) - \frac{1}{\alpha-1}$$

is increasing in $\alpha \in [1, 2]$

Proof. Define $f(\alpha) = \tilde{Q}_\alpha(A|B)$. From property 2 and property 3, we have

$$(\alpha-1) \frac{d}{d\alpha} f(\alpha) \geq f(\alpha) \log f(\alpha)$$

Then, we compute

$$\frac{d}{d\alpha} \frac{f(\alpha) - 1}{\alpha - 1} = \frac{(\alpha - 1) \frac{d}{d\alpha} f(\alpha) - (f(\alpha) - 1)}{(\alpha - 1)^2}$$

From the basic inequality of logarithm, we know $\log f(\alpha) \geq 1 - \frac{1}{f(\alpha)}$. Thus,

$$\frac{(\alpha - 1) \frac{d}{d\alpha} f(\alpha) - (f(\alpha) - 1)}{(\alpha - 1)^2} \geq \frac{f(\alpha) \log f(\alpha) - (f(\alpha) - 1)}{(\alpha - 1)^2} \geq 0$$

, which implies $\frac{f(\alpha) - 1}{\alpha - 1} = \frac{\tilde{Q}_\alpha(A|B) - 1}{\alpha - 1}$ is monotonic in α □

Then, we can make the following claim,

Proposition 3.

$$\frac{1}{\alpha - 1} \log \tilde{Q}_\alpha(A|B) \leq \frac{1}{\beta - 1} \tilde{Q}_\beta(A|B), \forall 1 < \alpha < \beta < 2$$

or equivalently,

$$\tilde{Q}_\alpha(A|B) \leq e^{\frac{\alpha - 1}{\beta - 1} \tilde{Q}_\beta(A|B)}$$

Proof. From the inequality $1 + rx \leq e^{rx}$ when $rx \geq 0$, it suffices to prove $\tilde{Q}_\alpha(A|B) \leq 1 + \frac{\alpha - 1}{\beta - 1} \tilde{Q}_\beta(A|B)$. From proposition 2, we know

$$\frac{1}{\alpha - 1} \tilde{Q}_\alpha(A|B) - \frac{1}{\alpha - 1} \leq \frac{1}{\beta - 1} \tilde{Q}_\beta(A|B) - \frac{1}{\beta - 1}$$

, which can be reduced into

$$\tilde{Q}_\alpha(A|B) \leq \frac{\beta - \alpha}{\beta - 1} + \frac{\alpha - 1}{\beta - 1} \tilde{Q}_\beta(A|B) \leq 1 + \frac{\alpha - 1}{\beta - 1} \tilde{Q}_\beta(A|B)$$

and completes the proof. □

Finally, we can make use of proposition 3 to finish our main problem by taking $\alpha \rightarrow 1$, which gives us

$$\text{Tr } A \left(\log \frac{A + B}{2} - \log B \right) \leq D(A|B) = \lim_{\alpha \rightarrow 1} \frac{1}{\alpha - 1} \log \tilde{Q}_\alpha(A|B) \leq \frac{1}{\beta - 1} \tilde{Q}_\beta(A|B), \forall \beta \in [1, 2]$$

Theorem 3.1.

$$\text{Tr } A (\log A + B - \log B) \leq \frac{1}{\alpha - 1} \text{Tr } (B^{\frac{1-\alpha}{2\alpha}} A B^{\frac{1-\alpha}{2\alpha}})^\alpha, \forall \alpha \in [1, 2]$$

Proof. Since $\text{Tr } A(\log A + B - \log B) - \log 2 = \text{Tr } A(\log \frac{A+B}{2} - \log B)$, we will show

$$\text{Tr } A(\log \frac{A+B}{2} - \log B) + \log 2 \leq \frac{1}{\alpha-1} \text{Tr } (B^{\frac{1-\alpha}{2\alpha}} AB^{\frac{1-\alpha}{2\alpha}})^{\alpha}$$

Similar to the previous argument, we need to show $2^{\alpha-1} \tilde{Q}_{\alpha}(A|B) \leq e^{\frac{\alpha-1}{\beta-1} \tilde{Q}_{\beta}(A|B)}$, for $1 < \alpha < \beta < 2$, which becomes

$$\tilde{Q}_{\alpha}(A|B) \leq e^{\frac{\alpha-1}{\beta-1} \tilde{Q}_{\beta}(A|B) - (\alpha-1) \log 2}$$

Observe

$$\tilde{Q}_{\alpha}(A|B) \leq \frac{\beta-\alpha}{\beta-1} + \frac{\alpha-1}{\beta-1} \tilde{Q}_{\beta} \leq 1 + \frac{\alpha-1}{\beta-1} \tilde{Q}_{\beta} - (\alpha-1) \log 2 \leq e^{\frac{\alpha-1}{\beta-1} \tilde{Q}_{\beta}(A|B) - (\alpha-1) \log 2}$$

The first inequality is the result of proposition 2. We can also directly get the second inequality due to the fact that $\log 2 \leq \frac{1}{\beta-1}$ for $1 < \beta < 2$.

Finally, take $\alpha \rightarrow 1$ we can finish the proof. $\square$

References

- [1] Müller-Lennert et al. *On Quantum Renyi entropies: A new generalization and some properties*. 2014.
- [2] Wilde et al. *Strong converse for the classical capacity of entanglement-breaking and Hadamard channels via a sandwiched renyi relative entropy*. 2014.
- [3] Leditzky. *Relative entropies and their use in quantum information theory*. 2016.